REAL-TIME DRIVER DROWSINESS DETECTION USING YOLOV8, HAAR CASCADES, AND CNN CLASSIFIERS

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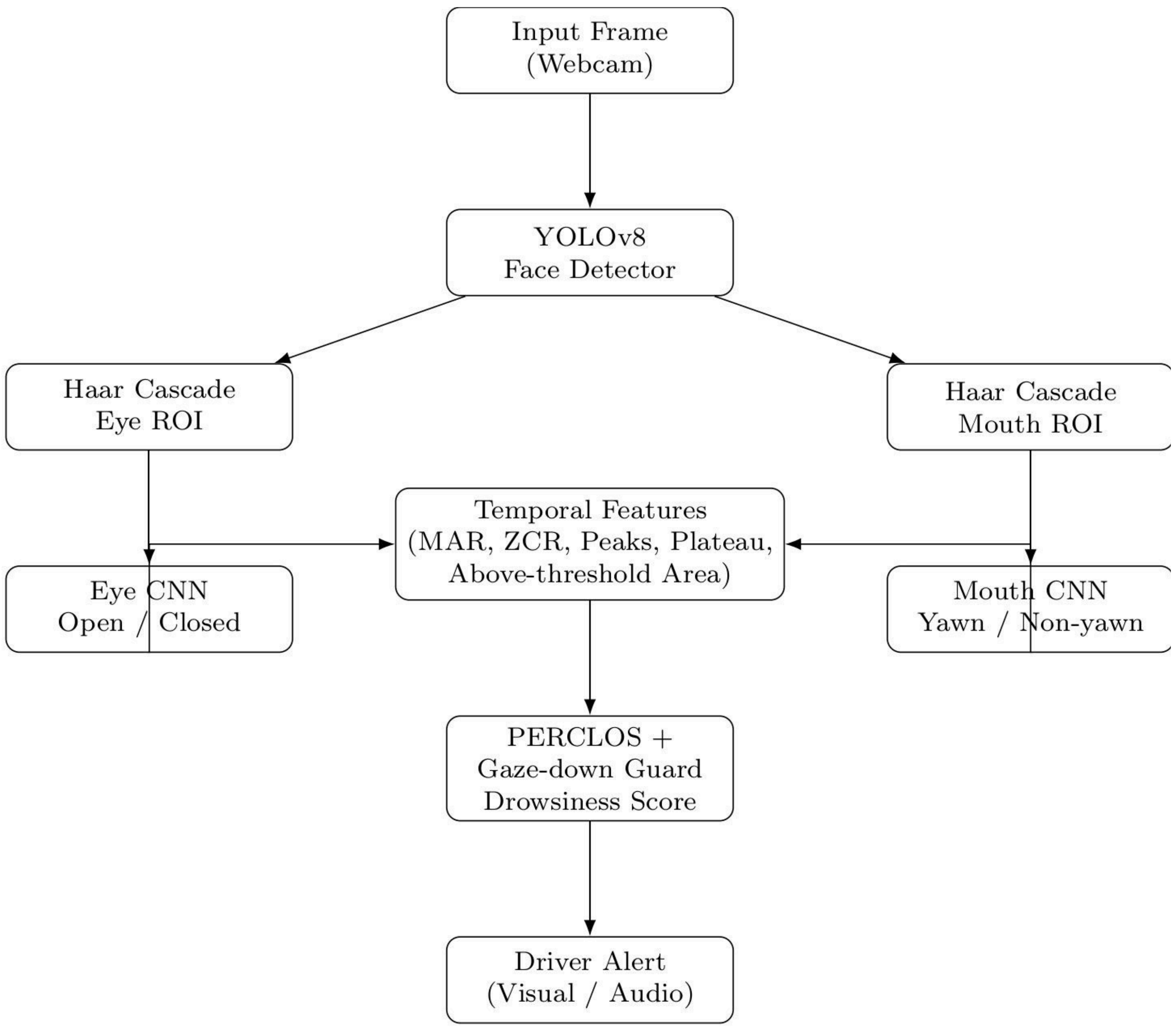
1. Abstract

Driver fatigue is a major factor in road accidents. We propose a real-time, vision-based system that uses YOLOv8 for robust face detection, Haar Cascades for region extraction, and lightweight CNNs for eye-state and yawn classification. The core innovation is fusing temporal features (MAR, PERCLOS) and a gaze-down guard for a stable, real-time drowsiness score on CPU-only hardware.

2. Model Architecture & Training Methodology

The proposed pipeline is a sequence of modular components that runs in real time.



The proposed real-time driver drowsiness detection pipeline (Figure 1) processes webcam frames using **YOLOv8** and **Haar Cascades** to get eye and mouth ROIs. These ROIs are classified by lightweight **CNNs** and the results are fused with **MAR, PERCLOS,** and **gaze-based temporal features**. This fusion generates a drowsiness score that triggers visual and audio alerts.

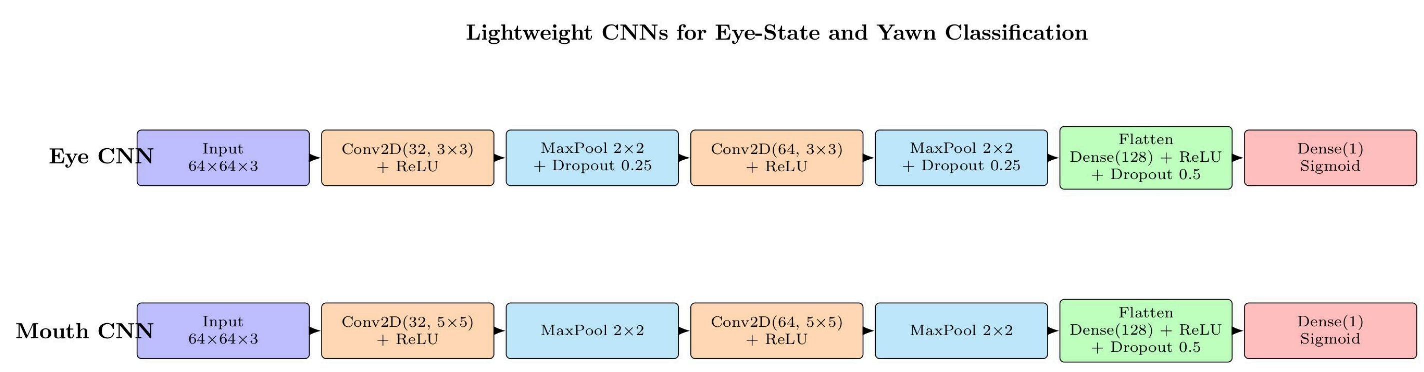


Figure 2. Internal structure of the lightweight CNNs used for eye-state and yawn classification.

The process begins with a YOLOv8 face detector analyzing the input frame. Haar Cascade classifiers then extract eye and mouth Regions of Interest (ROIs) from the detected face, which are fed into two lightweight CNNs for eye-state (open/closed) and yawn (yawn/non-yawn) classification.

For a stable drowsiness estimate, a temporal module aggregates these frame-level predictions, computing MAR-based features (zero-crossing rate, peaks, plateau duration, and above-threshold area) along with PERCLOS and a gaze-down guard. This final drowsiness score is then used to trigger a visual or audio driver alert in real time on a CPU-only laptop.

3.Testbed Setup

The system is evaluated on a standard laptop, USB webcam (640×480, 30 FPS), running YOLOv8, Haar Cascades,both CNNs on the CPU.

Our dataset includes ~80k eye images, ~5k mouth images, with an 80/20 train–validation split (16,190 eye and 1,023 mouth samples in validation)

We report eye-state, yawn training/validation accuracy, measure end-to-end FPS under typical indoor lighting conditions.

4.1 Training Results

The lightweight CNNs used an 80/20 train-validation split. Training accuracy for both models was high, around 97%. Validation accuracy was 86% for the Eye CNN (evaluated on 16,190 images) and 95% for the Mouth CNN (evaluated on 1,023 images) The Mouth CNN demonstrated stronger generalization.

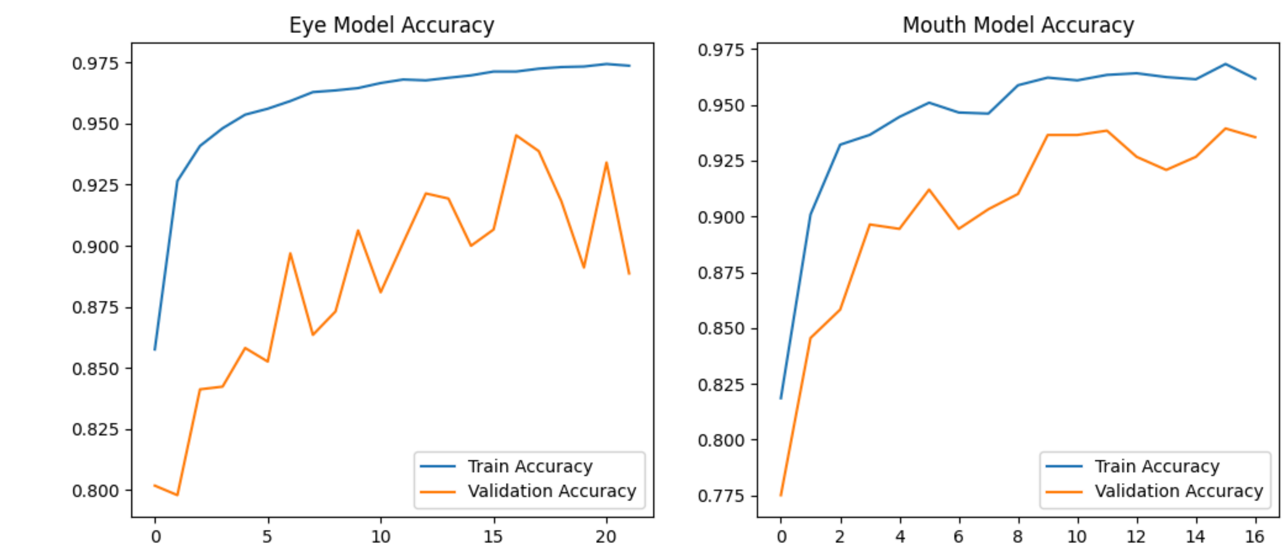


Figure 3. Training and validation accuracy of the eye-state CNN (left) and yawn CNN (right).

4.2 Evaluation Results

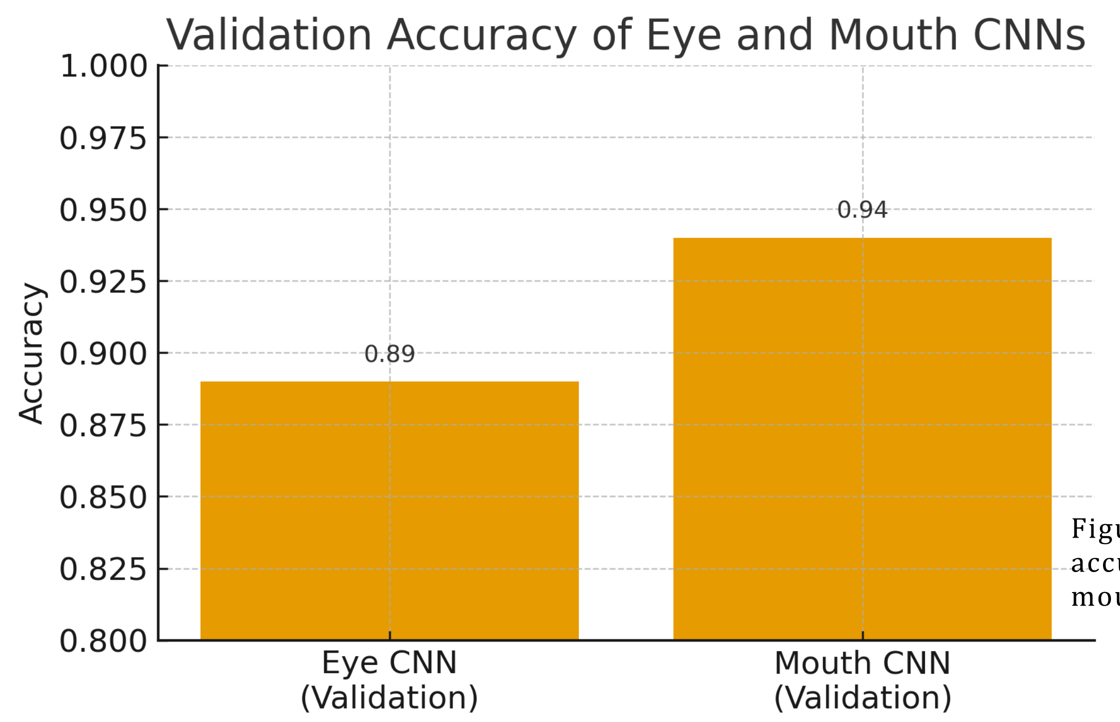


Figure 4. Validation accuracy of the eye and mouth CNNs.

5. Conclusion

The proposed system successfully combines modern deep learning (YOLOv8) with efficient classical components (Haar Cascades and lightweight CNNs) and temporal modeling.

- Achieved **real-time performance** (Avg. 15.7 FPS) on a **CPU-only laptop**.
- Demonstrated **high classification accuracy** for both eye-state (86%) and yawn (95%).
- The temporal features and PERCLOS-based decision module enable **robust and reliable** fatigue prediction.